secpod

Annual VULNERABILITY REPORT 2024

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2024 Annual Vulnerability Report

What did 2024 have in store for the world of cybersecurity?

As we reflect on the year, it's clear that the threat landscape has evolved in ways we could not have predicted. With an overwhelming increase in global vulnerabilities, organizations faced unprecedented challenges in safeguarding their IT networks. In 2024, **a staggering 40704 vulnerabilities were identified**—marking a continued upward trend from the previous year. This increase in vulnerabilities, including a notable rise in zero-day threats, has put immense pressure on IT security teams worldwide.

What does this mean for organizations moving forward? As vulnerabilities continue to multiply at an alarming rate, IT teams must remain vigilant and proactive in their approach to cybersecurity. The security risks of 2024 demanded an even more robust and agile response, highlighting the importance of timely and effective patching strategies.

In this Annual Vulnerability Report, we dive into the key vulnerabilities that shaped the year and grabbed the attention of security experts and organizations alike. From critical zero-days to top affected vendors, this report provides you with essential insights into the vulnerabilities that require your immediate focus. Whether you're already addressing some of these issues or need to take swift action, securing your network against the growing tide of threats remains more crucial than ever.

As we move into the future, one thing is clear: the journey to secure our digital landscapes is ongoing, and staying ahead of emerging vulnerabilities is the key to protecting your organization from the next wave of attacks.

What does the Report consist of?

The report consist of the essential details of the vulnerabilities that were discovered in the year 2024. Note that the vulnerability details are researched and mentioned in the report based on the publish date.

You will find detailed insights on the following:

- Key Findings from SecPod's Security Research Team
- Total number of vulnerabilities
- Vulnerability Distribution based on CVSS v3 &v4 Algorithm, Exploitability Score, and Impact score
- Top 10 Affected Vendors/products
- Top 10 Affected Operating systems
- Top 10 Affected Applications
- Top 10 Affected Hardware
- Top 10 most critical vulnerabilities
- Analytics of Malware Vulnerability Enumeration (MVE)
- Vulnerability Prediction of the upcoming months using the ARIMA model

Key Findings from SecPod's Security Research Team

- 40704 is the total number of vulnerabilities discovered in 2023, 30% more than the 31279 vulnerabilities discovered in 2023.
- As per CVSS v3 & CVSS v4, 19100 vulnerabilities were reported with critical & high severity in the year 2024.
- 31 zero days were reported in 2024.
- 129 of the total vulnerabilities discovered in 2024 are widely exploited.
- 23 of the total vulnerabilities discovered in 2024 have public exploits available.
- 120 of the total vulnerabilities discovered in 2024 are Malware Exploiting Vulnerabilities, causing High-fidelity attacks.
- 429 web browser vulnerabilities were discovered in 2024.

Vulnerability Trend 2024

Total number of vulnerabilities: **40704**

Vulnerabilities with critical & high severity as per CVSS v3 & v4: **19100**

Widely Exploited Vulnerabilities: **129**

Vulnerabilities with Public Exploit available: **23**

Vulnerabilities causing high-fidelity attacks: **120**

SecPod's Security Intelligence Coverage in 2024

Advanced Vulnerability Coverage

- Total No of CVEs Covered: 15575
- No of Local Checks: 24263
- No. of Remote Checks: 654
- Zero-day CVEs covered: 31/25

CVE Coverage based on Platforms

- Windows - 3368
- Linux - 12630
- macOS – 1904
- Common Remediation Enumeration Coverage (CRE): 2955
- Network Device Vulnerabilities: 654
- Total No of Misconfigurations Covered (CCE): 2408
- Total No of Patches Covered: 2440
- Total No of Third-party applications Patches Covered: 1292
- Total No of Misconfigurations Patches Covered: 1148

Compliance Benchmark Coverage in 2024

- Windows Server 2019 STIG Coverage
- Ubuntu 23.04 benchmarks revised
- Windows Server 2022 STIG Coverage
- Windows 11 CIS Coverage
- Windows 10 CIS Coverage
- Windows Server 2016 CIS Coverage
- Windows Server 2019 CIS Coverage
- Windows Server 2022 CIS Coverage
- MySQL 8 General Compliance
- Oracle DB 19C General Compliance
- PostGRESQL 16 General Compliance
- MSSQL 2022 General Compliance
- Mac OS 15 General Compliance
- Amazon Linux 2023 General Compliance
- MSSQL 2019 General Compliance
- MSSQL 2017 General Compliance
- MSSQL 2016 General Compliance

Monthly Number of Vulnerabilities Discovered

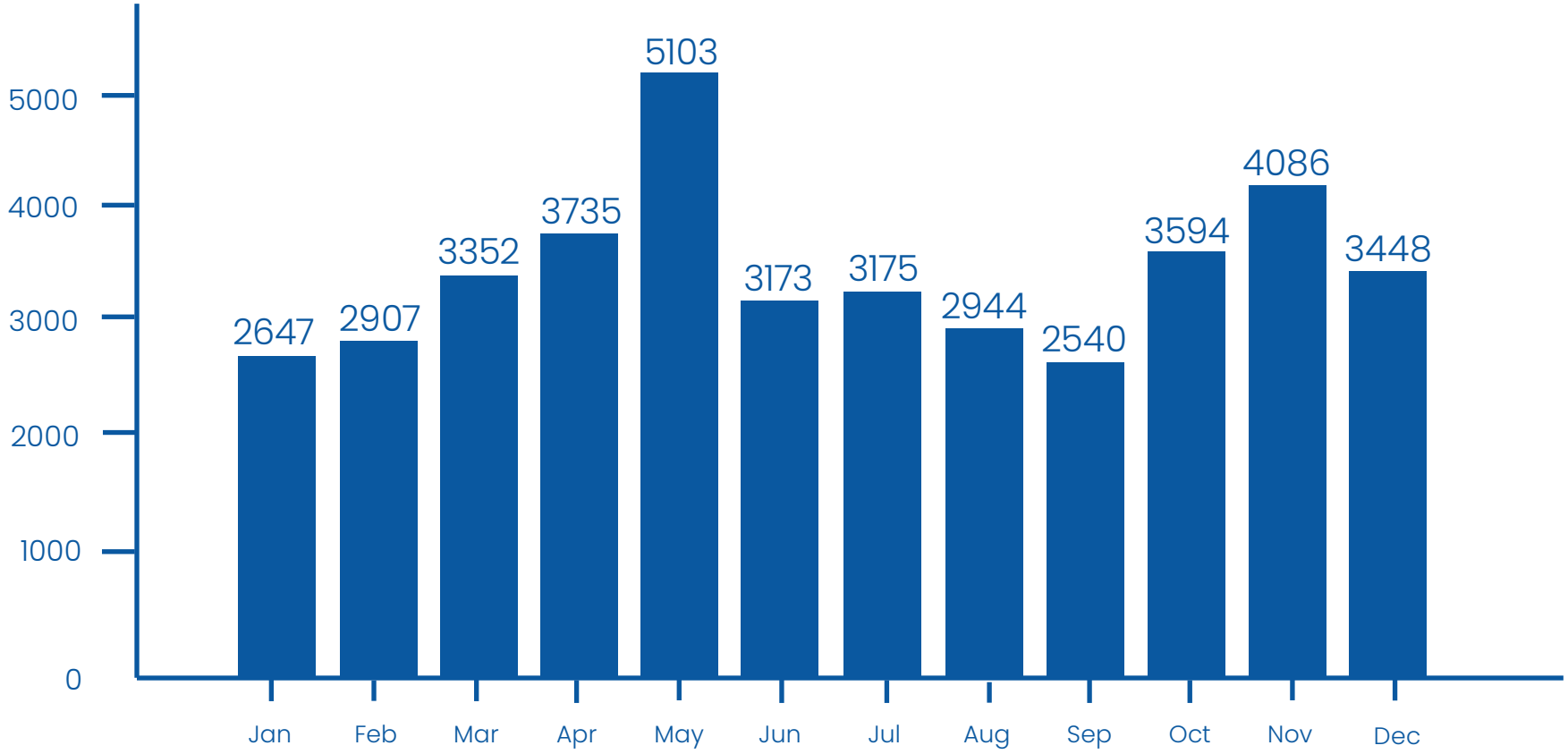


Figure 1: Shows the number of vulnerabilities published monthly in 2024

The highest number of vulnerabilities was detected in May.

Vulnerability Severity Distribution based on CVSS v3 & CVSS v4

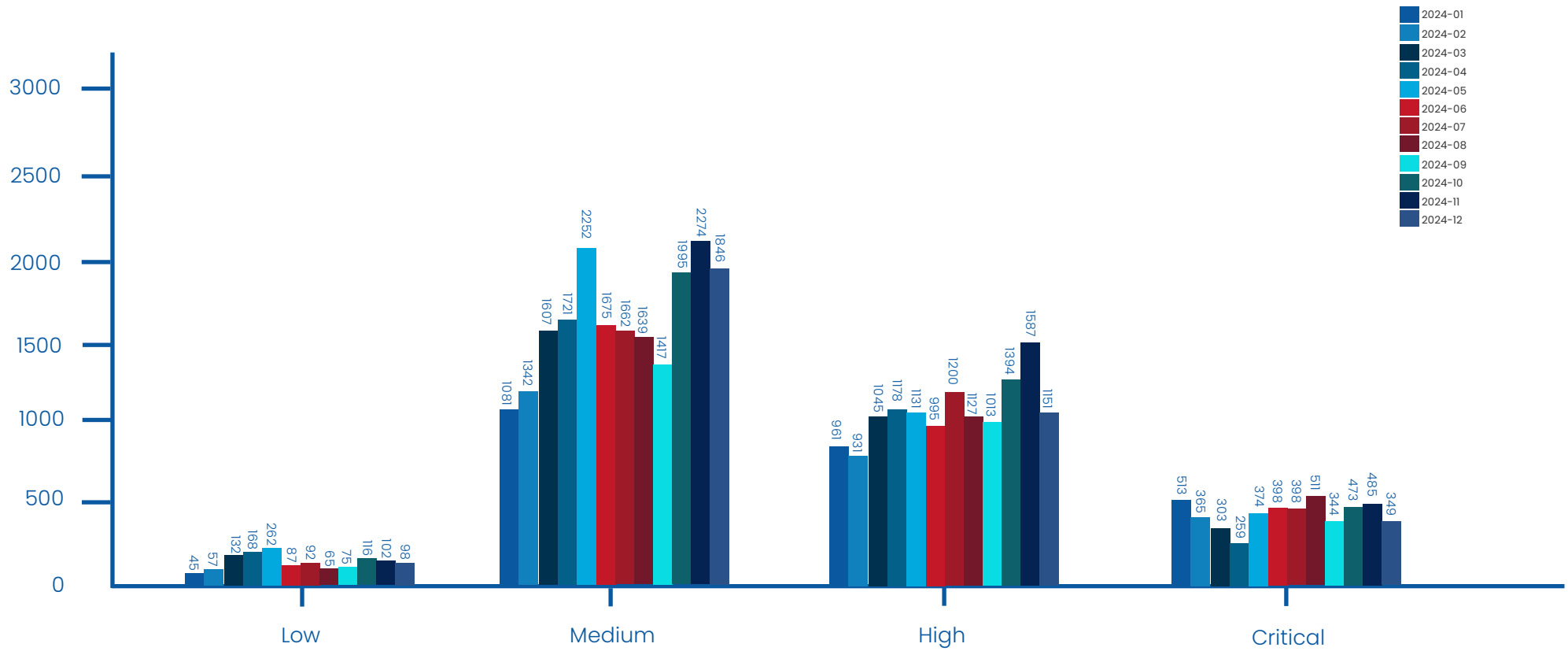


Figure 2: Depicts the vulnerability severity distribution based on CVSS v3 & CVSS v4

According to the CVSS v3 and v4 scores, 4750 vulnerabilities are reported as critical, 14370 as high severity, 20427 as medium severity, and 1299 as low severity.

Top 10 Affected Vendors/Products

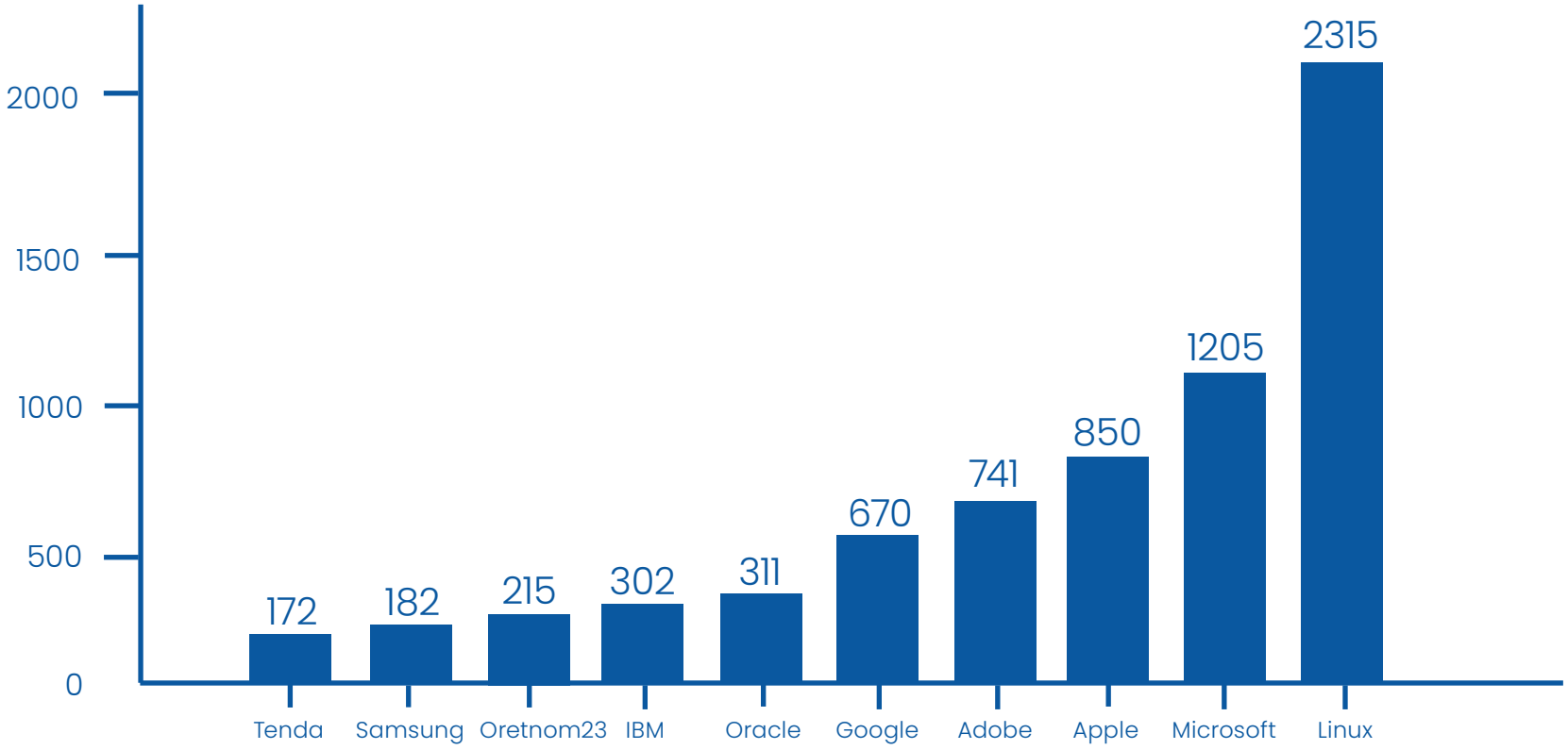


Figure 5: Shows the Top 10 vendors affected by CVEs

Linux and Microsoft are the most affected vendors in 2024. Respectively, they have reported 2315 and 1205 vulnerabilities each. Linux being the top affected is unsurprising given the widespread use of Linux in servers, cloud environments, IoT devices, and as the foundation for many enterprise and consumer technologies. Its open-source nature also encourages comprehensive vulnerability disclosure and tracking.

Top 10 Affected Operating Systems

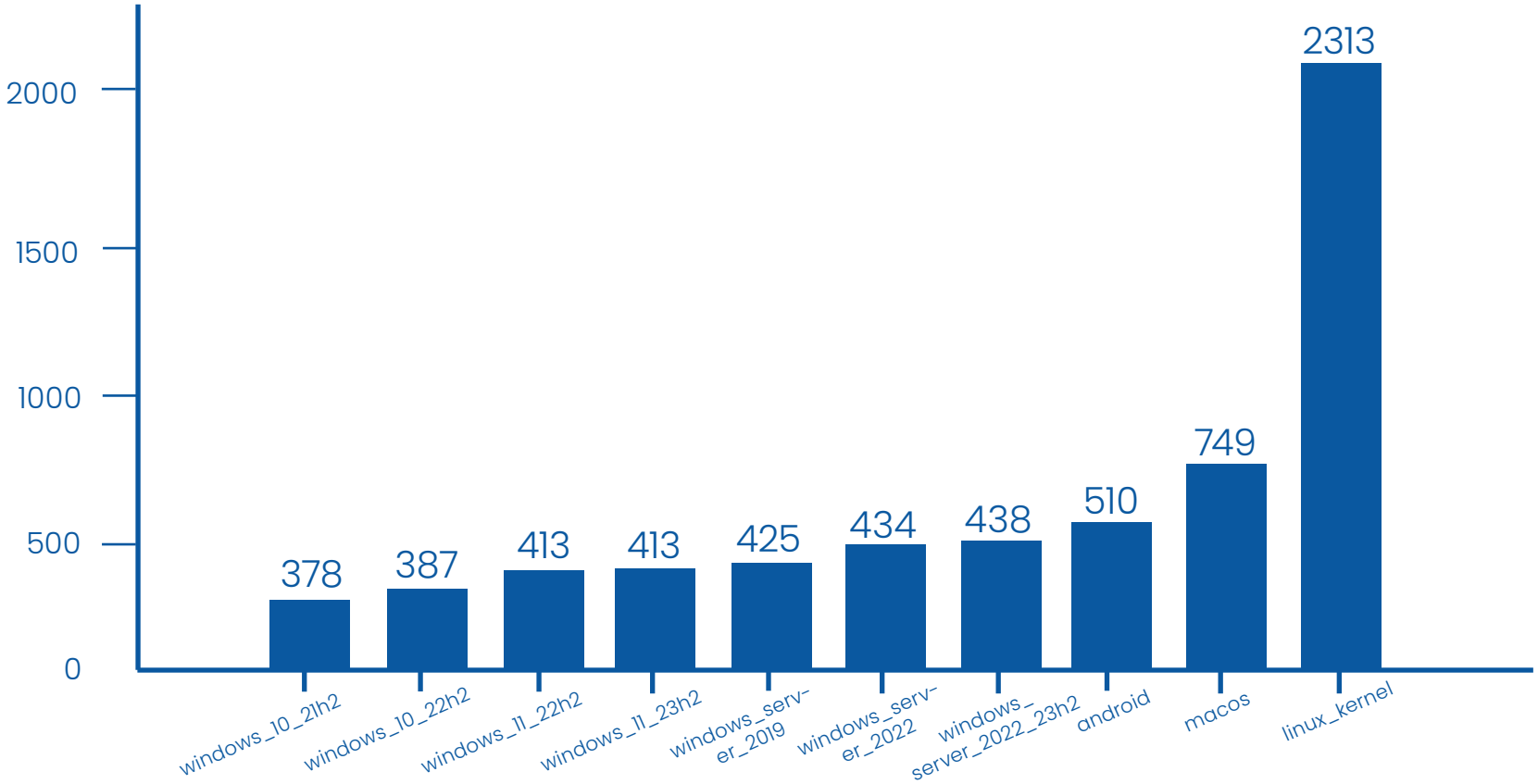


Figure 6: Shows the Top 10 Operating Systems Affected by CVEs

Linux is the most affected operating system, with a total of 2313 vulnerabilities, followed by macOS, with 749 vulnerabilities.

Top 10 Affected Applications

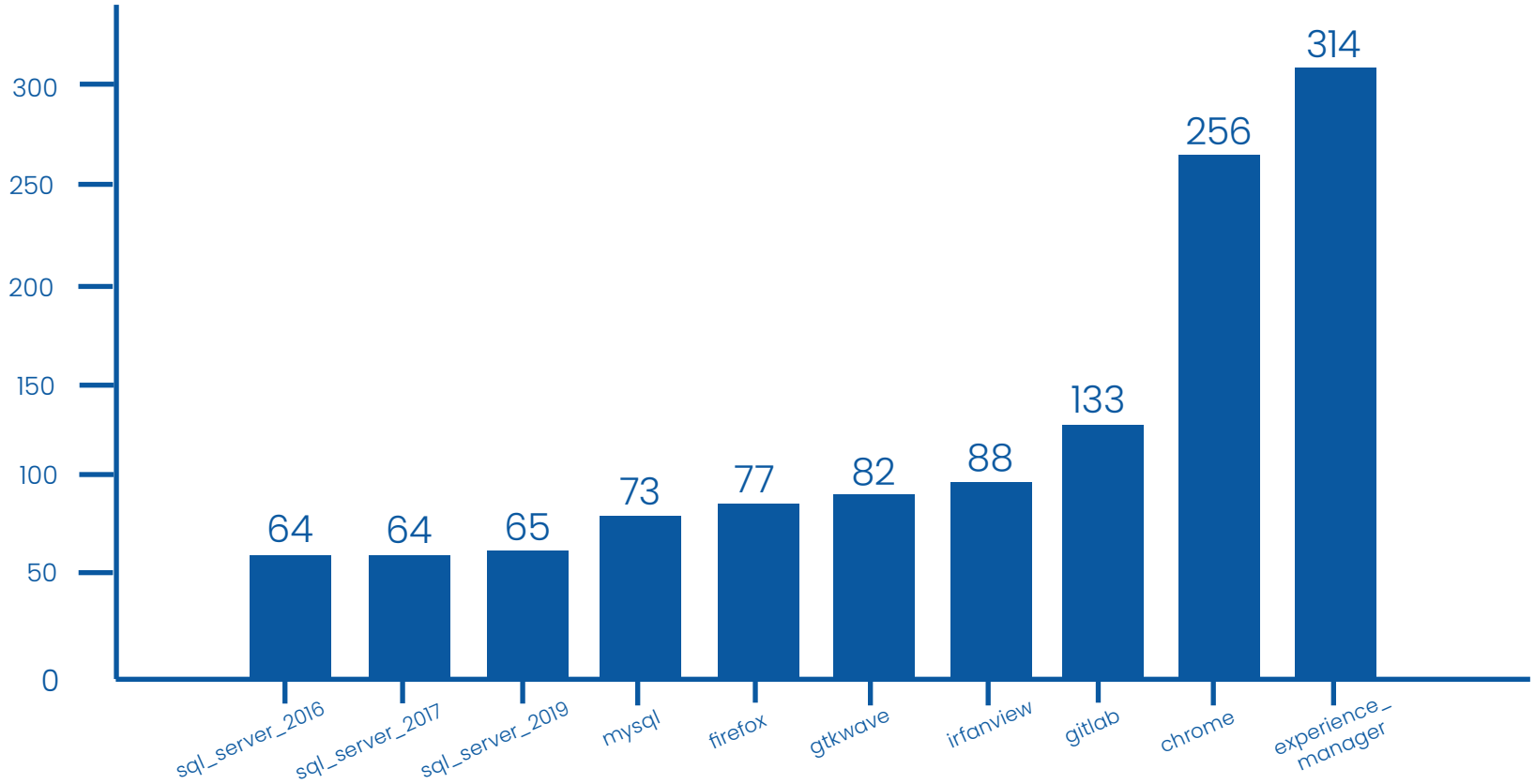


Figure 7: Show the Top 10 Applications Affected by CVEs

Experience Manager was the most affected application in 2024, with a total of 314 vulnerabilities. Adobe Experience Manager is a leading content management system (CMS) and digital experience platform. Enterprises widely use it to manage web content, digital assets, and customer experiences across multiple channels, including websites, mobile apps, and social media.

Top 10 Affected Hardware

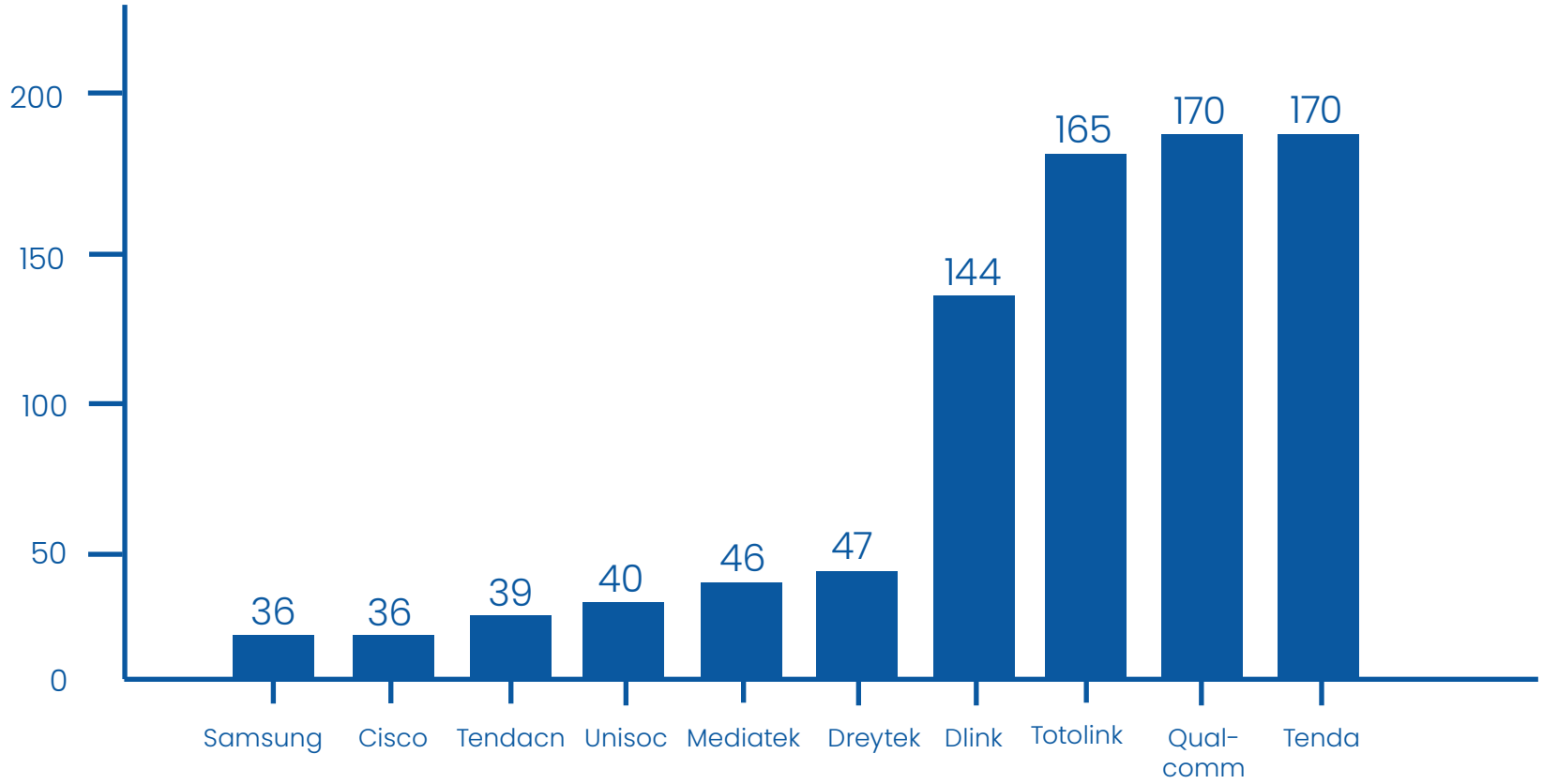


Figure 8: Shows the Top 10 Hardware Affected by CVEs

Tenda and Qualcomm are the most affected hardware in 2024, with a total of 170 vulnerabilities each.

Top 20 Most Critical Vulnerabilities

This section details the Top 20 most critical vulnerabilities discovered in 2024. The information on the vulnerabilities includes the CVE details, CVSS number, the affected products, and the impact of the vulnerability. We recommend you immediately identify and remediate these vulnerabilities in your network to prevent potential attacks.

S.No.	CVE ID	AFFECTED PRODUCTS	CVSS SCORE	IMPACT
1	CVE-2024-23897	Jenkins integrated command line interfaces (CLI)	9.8	Read arbitrary files on the Jenkins controller file system
2	CVE-2024-0204	Fortra's GoAnywhere MFT	9.8	Unauthorized user can create an admin user via the admin portal
3	CVE-2024-21410	Microsoft Exchange Server	9.8	Gain privileges as the victim client and to perform operations on the Exchange server on the victim's behalf
4	CVE-2024-21413	Microsoft Outlook	9.8	Gain high privileges, which include read, write, and delete functionality

S.No.	CVE ID	AFFECTED PRODUCTS	CVSS SCORE	IMPACT
5	CVE-2024-3094	XZ Utils	10	Any software linked against this library can be used for intercepting and modifying the data interaction with this library
6	CVE-2024-1709	ConnectWise ScreenConnect	10	Direct access to confidential information or critical systems
7	CVE-2024-42789	Fortinet FortiOS	9.8	Execute unauthorized code or commands via specially crafted HTTP requests
8	CVE-2024-3400	Palo Alto Networks PAN-OS	10	Execute arbitrary code with root privileges on the firewall
9	CVE-2024-4358	Telerik Report Server	9.8	Gain access to Telerik Report Server restricted functionality via an authentication bypass vulnerability
10	CVE-2024-30080	Microsoft Windows	9.8	Remote code execution on the server side

S.No.	CVE ID	AFFECTED PRODUCTS	CVSS SCORE	IMPACT
11	CVE-2024-4879	ServiceNow Now Platform	9.3	Remote Code Execution leading to compromise, data theft, and disruption of business operations
12	CVE-2024-38077	Microsoft Windows	9.8	Remote Code Execution leading to complete system compromise, data theft, and unauthorized access to sensitive information
13	CVE-2024-38063	Microsoft Windows	9.8	Remote Code Execution allowing attackers to install programs, view, change, or delete data, or create new accounts with full user rights
14	CVE-2024-36401	GeoServer	9.8	Remote Code Execution potentially leading to full server compromise
15	CVE-2024-38178	Microsoft Windows	9.8	Remote Code Execution leading to system compromise
16	CVE-2024-4577	PHP	9.8	Unauthorized command execution on server, Potential full system compromise, Deploying ransomware

S.No.	CVE ID	AFFECTED PRODUCTS	CVSS SCORE	IMPACT
17	CVE-2024-6678	GitLab	9.9	Remote Code Execution to gain access to a self-hosted GitLab instance, if configured to provide authentication using SAML
18	CVE-2024-9680	Mozilla Firefox	9.8	Arbitrary code execution in the content process by exploiting a use-after-free in Animation timelines
19	CVE-2024-7589	VMware vCenter Server	9.8	Remote code execution via a flaw in the DCERPC protocol
20	CVE-2024-43639	Microsoft Windows	9.8	Specially crafted application can leverage a cryptographic protocol vulnerability in Windows Kerberos to perform remote code execution against the target

Zero Days Coverage from SecPod

This section contains details of the Zero-day CVEs discovered in 2024

S.No.	CVE ID	AFFECTED PRODUCTS	CVSS SCORE	IMPACT
1	CVE-2024-0519	Google Chrome	8.8	Memory Corruption
2	CVE-2024-23222	Apple WebKit	8.8	Memory Corruption
3	CVE-2024-23225	Apple iOS	7.8	Memory Corruption
4	CVE-2024-23296	Apple iOS	7.8	Memory Corruption
5	CVE-2024-29745	Google Android	5.5	Information Disclosure
6	CVE-2024-29748	Google Android	7.8	Privilege Escalation
7	CVE-2024-4671	Google Chrome	9.6	Security bypass

S.No.	CVE ID	AFFECTED PRODUCTS	CVSS SCORE	IMPACT
8	CVE-2024-4947	Google Chrome	8.8	Remote code execution
9	CVE-2024-30051	Microsoft Windows	7.8	Privilege Escalation
10	CVE-2024-5274	Google Chrome	8.8	Remote code execution
11	CVE-2024-4610	ARM Android	5.5	Privilege Escalation
12	CVE-2024-32896	Google Android	7.8	Privilege Escalation
13	CVE-2024-38030	Microsoft Windows	7.8	Privilege Escalation
14	CVE-2024-36971	Google Android	7.8	Memory Corruption
15	CVE-2024-38178	Microsoft Windows	7.5	Memory Corruption
16	CVE-2024-38106	Microsoft Windows	7	Privilege Escalation
17	CVE-2024-38193	Microsoft Windows	7.8	Memory Corruption

S.No.	CVE ID	AFFECTED PRODUCTS	CVSS SCORE	IMPACT
18	CVE-2024-38107	Microsoft Windows	7.8	Memory Corruption
19	CVE-2024-38189	Microsoft Project	8.8	Remote Code Execution
20	CVE-2024-7971	Google Chrome	8.8	Memory Corruption
21	CVE-2024-7965	Google Chrome	8.8	Memory Corruption
22	CVE-2024-21338	Microsoft Windows	7.8	Privilege Escalation
23	CVE-2024-4671	Google Chrome	8.8	Memory Corruption
24	CVE-2024-43047	Qualcomm DSP Service	7.8	Memory Corruption
25	CVE-2024-44068	Samsung Exynos DSP	8.1	Privilege Escalation
26	CVE-2024-44308	Apple WebKit	8.8	Memory Corruption
27	CVE-2024-44309	Apple WebKit	6.1	Remote code execution

S.No.	CVE ID	AFFECTED PRODUCTS	CVSS SCORE	IMPACT
28	CVE-2024-49039	Microsoft Windows	8.8	Privilege Escalation
29	CVE-2024-49138	Microsoft Windows	7.8	Privilege Escalation
30	CVE-2024-7262	Kingsoft WPS Office	7.8	Privilege Escalation
31	CVE-2024-9680	Mozilla Firefox	9.8	Remote code execution

Top 20 Misconfigurations

CCE-97142-4
CCE-98055-7
CCE-95633-4
CCE-95066-7
CCE-43114-8
CCE-96684-6
CCE-99759-3
CCE-44884-5
CCE-97667-0
CCE-100094-2
CCE-100076-9
CCE-37441-3
CCE-100084-3
CCE-9036-5
CCE-35277-3
CCE-10640-1
CCE-100118-9
CCE-100107-2
CCE-99444-2
CCE-95517-9

Analysis of High-Fidelity Attacks

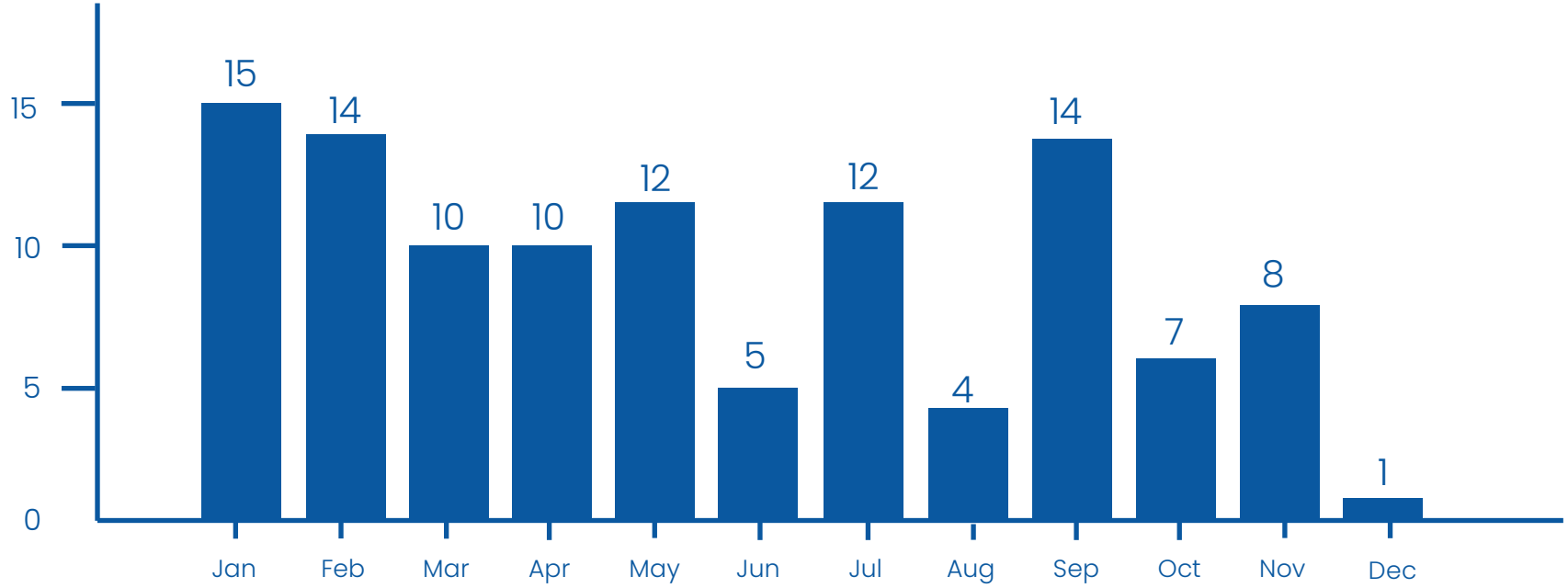


Figure 9: Depicts the monthly number of vulnerabilities that cause high-fidelity attacks

At SecPod, we go through all the discovered CVEs and cross-reference them with our in-depth Malware Vulnerability Enumeration (MVE) data. This helps us pinpoint vulnerabilities that are vulnerable to high-fidelity attacks. In 2024, we found 120 such vulnerabilities that could be exploited with high precision. It's crucial to act quickly and fix these vulnerabilities to protect your network from cyberattacks.

Vulnerability Prediction 2024

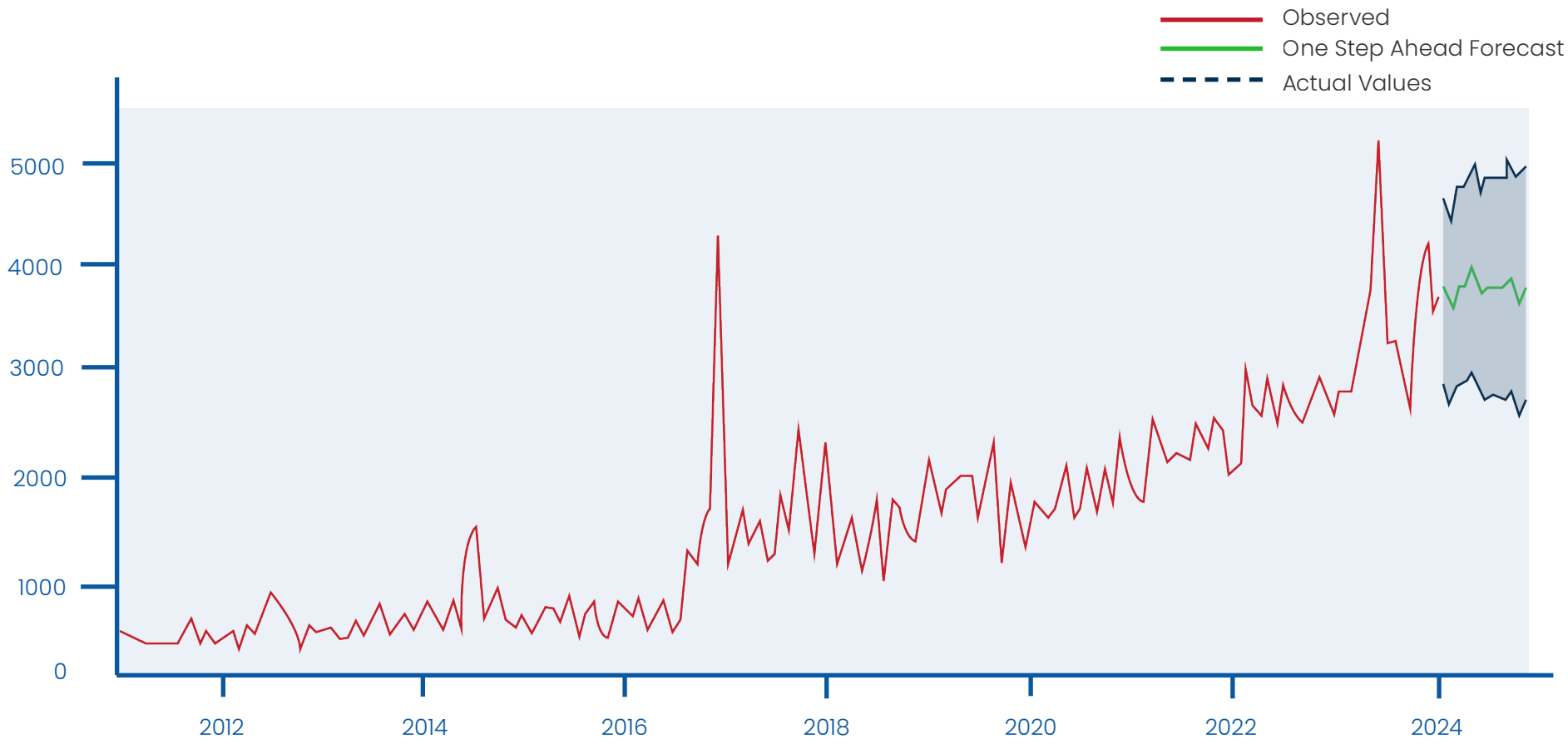
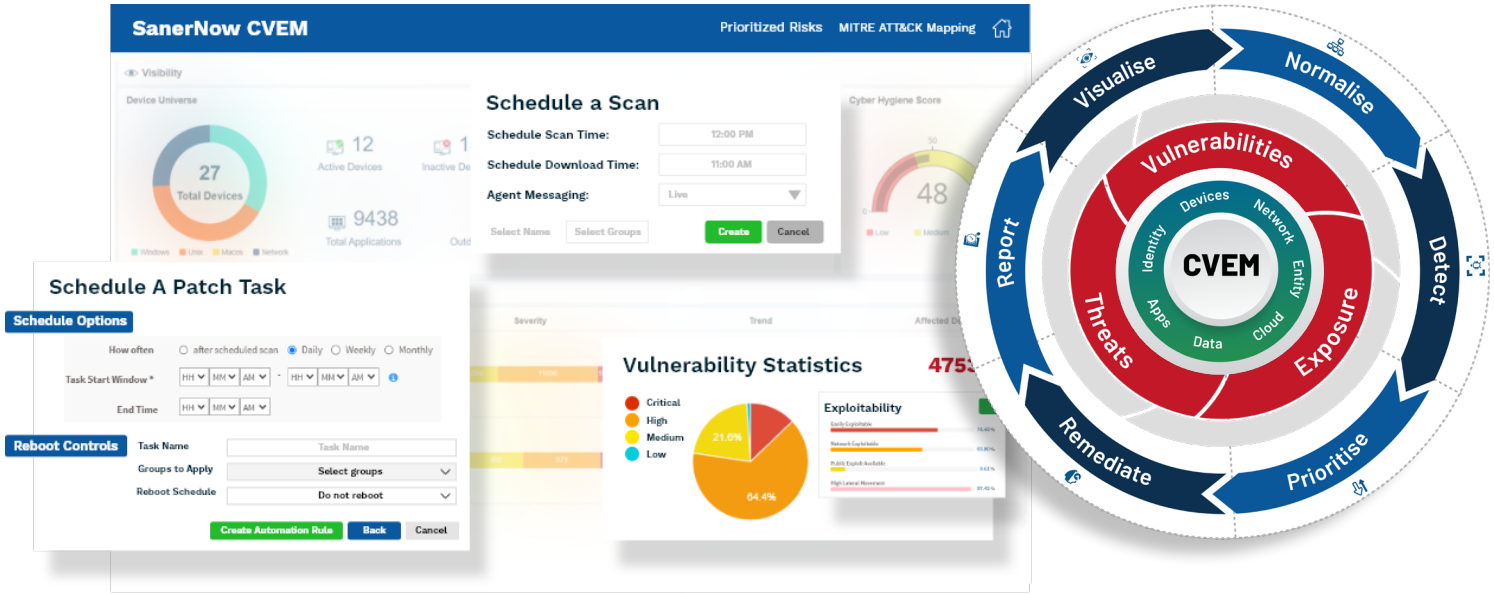


Figure 10: Forecasting Number of Vulnerabilities year-wise

By observing the vulnerability trend over the years from SecPod, we predict over 44,000 vulnerabilities in 2025. This prediction is based on the ARIMA (Autoregressive Integrated Moving Average) model.

Stay Ahead and Secure with SanerNow Continuous Vulnerability and Exposure Management

SecPod SanerNow Continuous Vulnerability and Exposure Management offers a continuous, automated, and truly integrated solution to discover, prioritize, and remediate vulnerabilities and numerous other security risks from a single centralized console. SanerNow's ability to discover vulnerabilities and exposures beyond CVEs, homegrown vulnerability intelligence library with 190,000+ vulnerability checks, rapid scanning techniques, integrated patching, CISA-SSVC-based prioritization, and remediation controls elevates the whole vulnerability and exposure management process. SanerNow provides all that you need to secure your organization from exposures, vulnerabilities, and misconfigurations and prevent a wide range of cyberattacks.



SecPod is a cyber security technology company with a mission to prevent cyberattacks on organizations. Our Advanced Vulnerability Management platform helps implement cyber hygiene measures, making it more difficult for attackers to access systems and companies' vital information.

SecPod SanerNow is a Cyber Hygiene platform providing continuous visibility to IT infrastructure. It identifies vulnerabilities, misconfigurations, and security risk exposures, mitigates loopholes to reduce the attack surface, measures compliance, and helps automate remediation. Our product philosophy is offering an easy-to-use solution with fast time-to-value that improves an organization's IT risk posture at a lower total cost of ownership vs. using point solutions.

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